

August 4, 2026

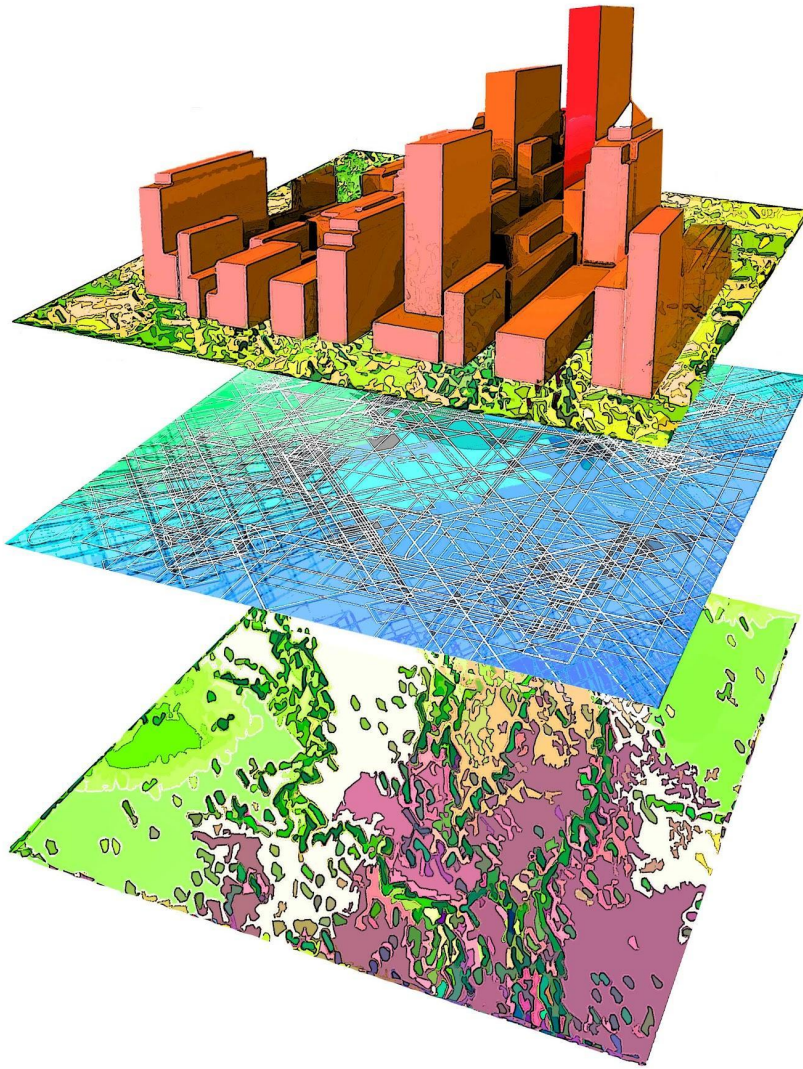
Black Women's Cambridge

A Case Study on Critical GIS and Black Feminist Geographies

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Geographic Information Systems (GIS)

Computer programs that **store, manage, analyze, and visualize** geographic data



Generalizes Knowledge

Seeks to generate “universally applicable principles or to make generalizations about the world”

Imposes Binaries

Positions one interpretation/representation as “unquestionable facts” and classifies data to fit this narrative

Purports Objectivity and Neutrality

All data and visualizations are objective and neutral

Critical GIS

A set of **practices, methods, and approaches to GIS** that push back against its perceived objectivity and positivist assumptions.

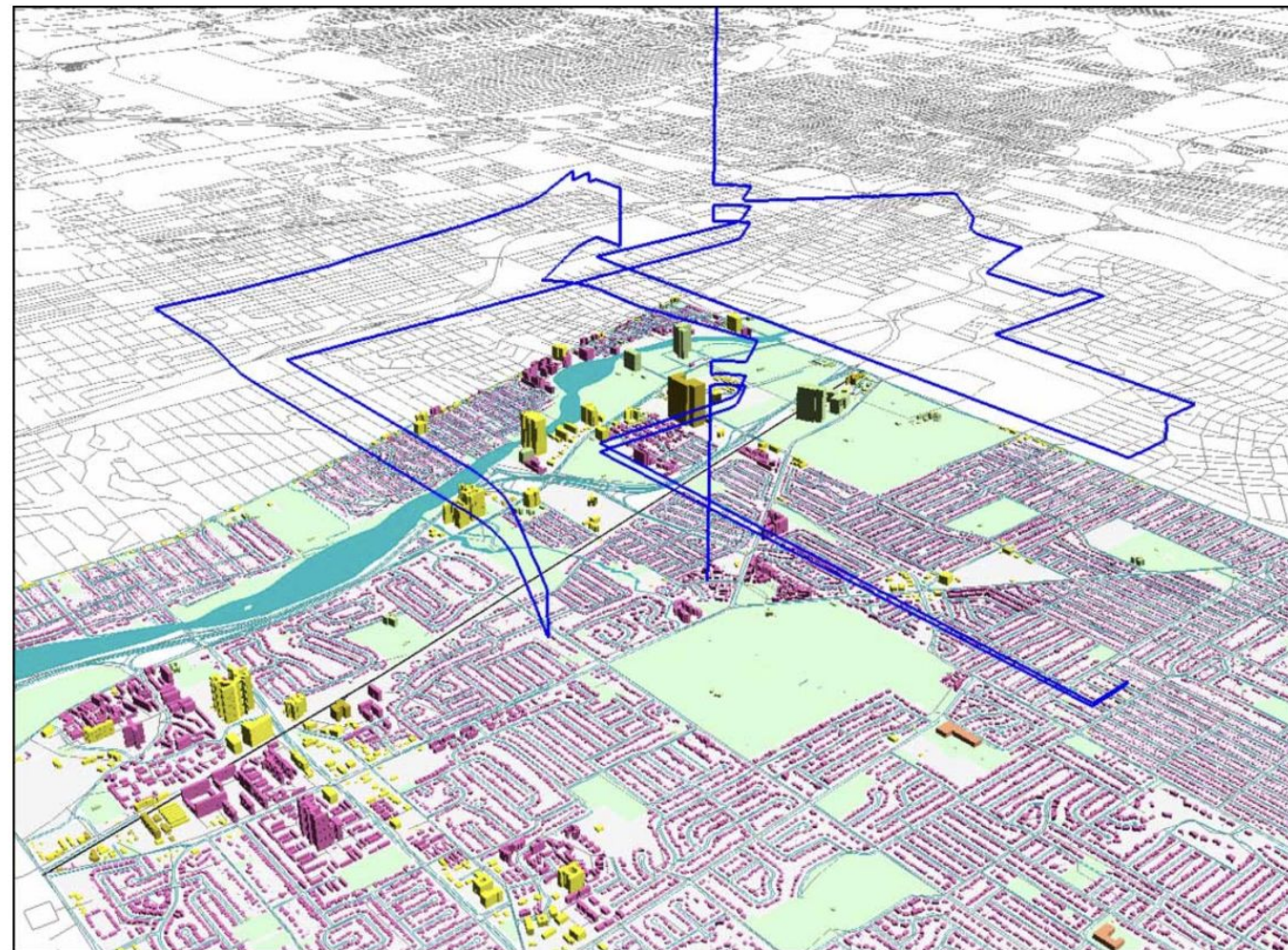


Figure 1 Nada's life path on a typical weekday before September 11.

Situated Knowledge

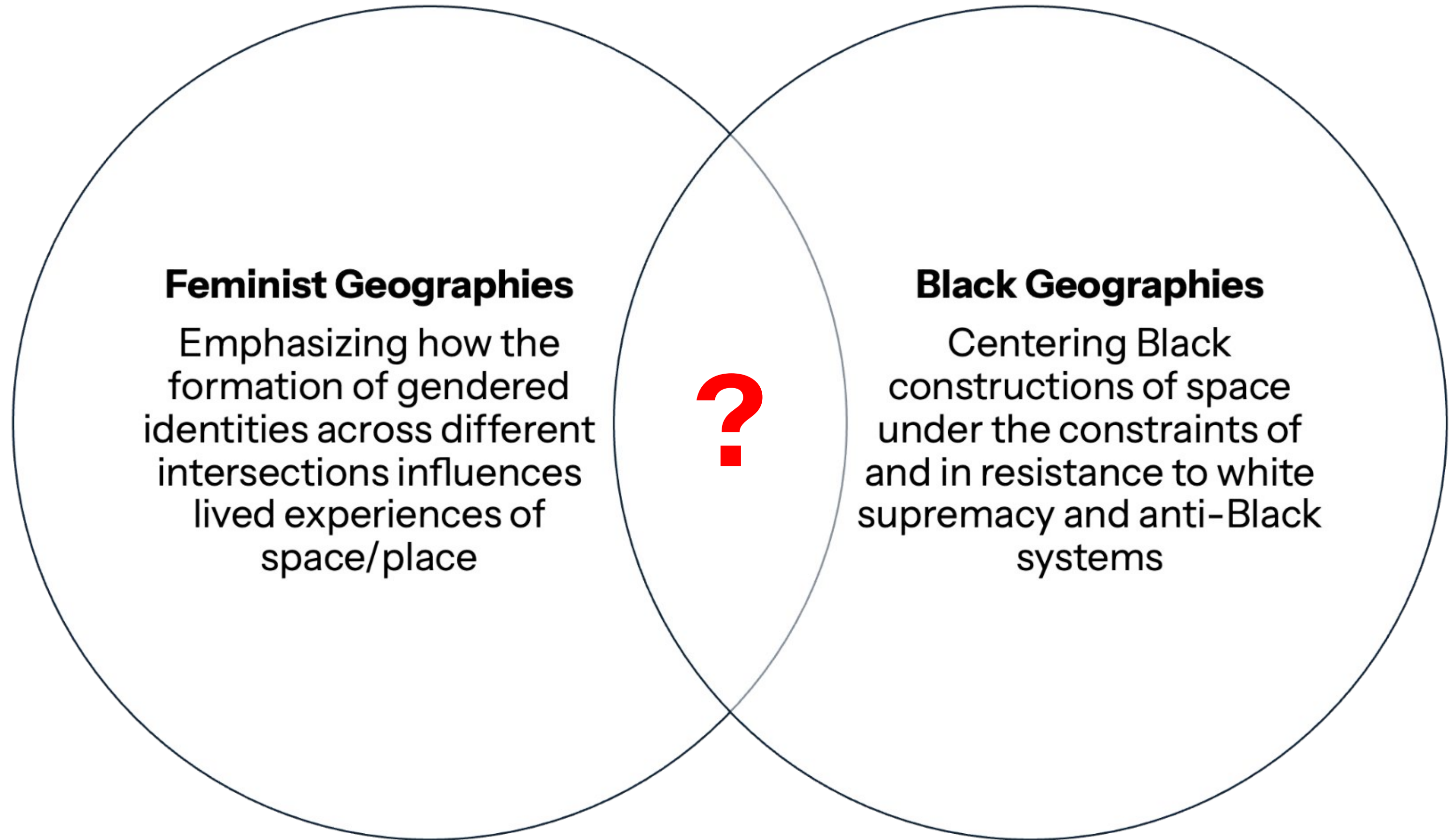
Acknowledges that “all knowledge comes from somewhere and is therefore partial”

Multiple Perspectives

Examines knowledge across multiple scales and contexts

Bodies and Emotions

Requires subjectivity by “recognizing emotion and embodiment as ways of knowing”

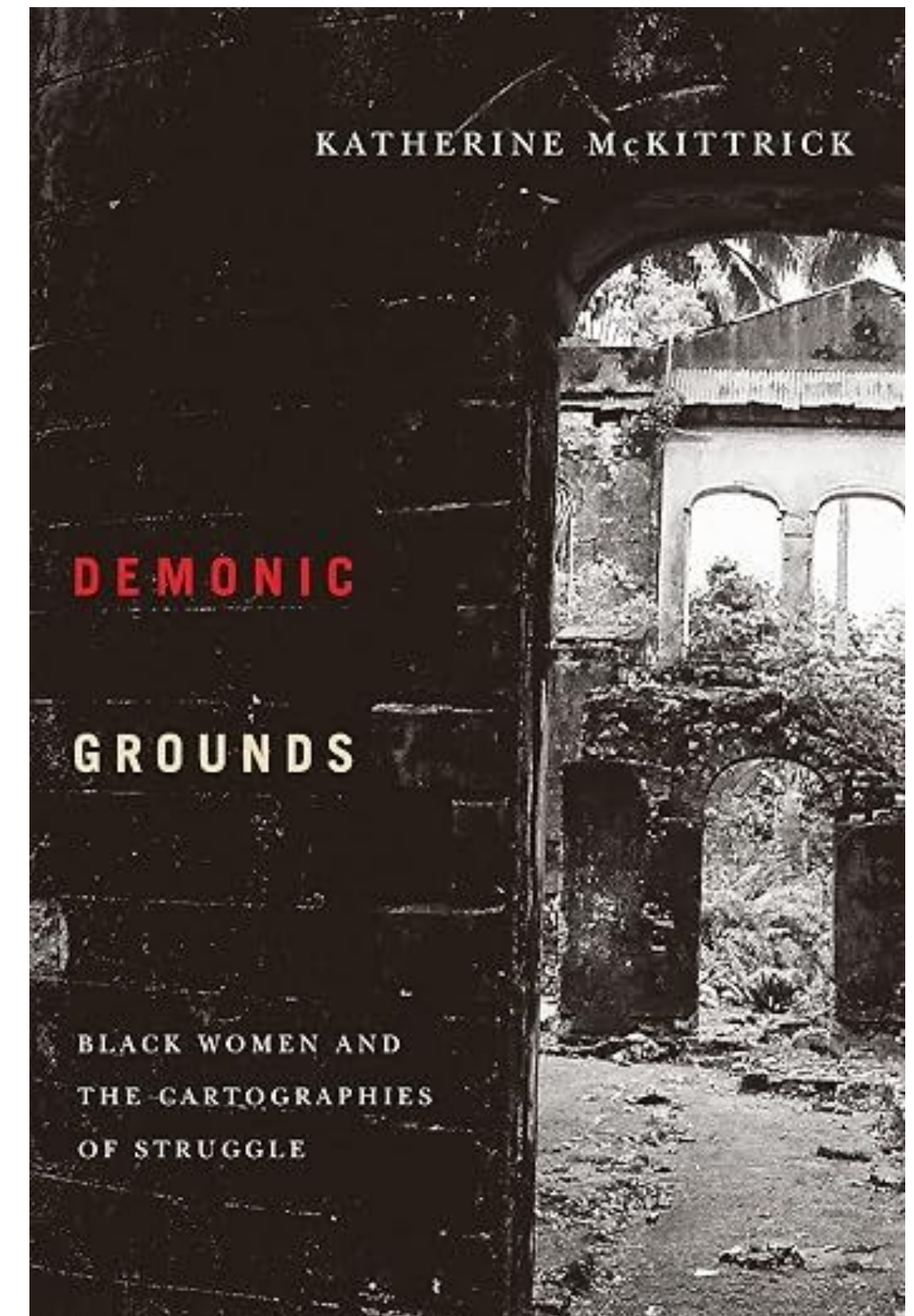


Black Feminist Geographies

Black feminist geographies describe how Black women's uniquely intersectional experience of racial-sexual hierarchies and oppression shape our creation and definition of geographic landscapes.

This creates an ambiguous, **subjective experience of geography** that captures understandings of space made “both *for* [us] through processes of control and *by* [us] through means of resistance.”

However, these geographies have not been directly considered or applied to critical GIS.



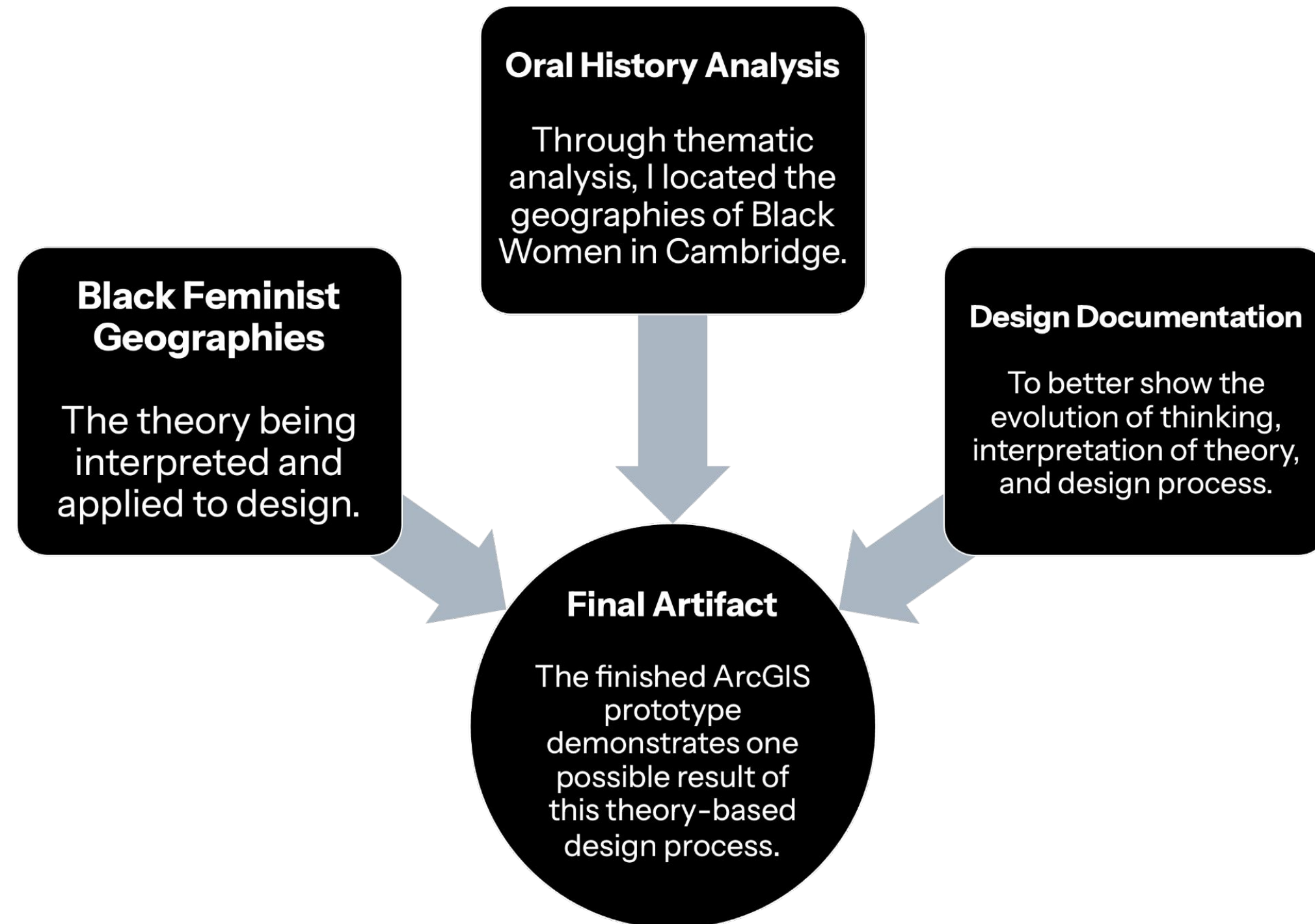
Demonic Grounds: Black Women and the Cartographies of Struggle by Katherine McKittrick

Inserting Black feminist perspectives into critical GIS

How might visualizing Black feminist geographies disrupt traditional assumptions in GIS technology?

Design as a Mode of Inquiry

A **research-through-design study** of Black feminist geographies using **oral histories** and **ArcGIS** to visualize Black women's geographies of Cambridge.



Oral History Thematic Analysis

Common Neighborhoods, Places, and Streets

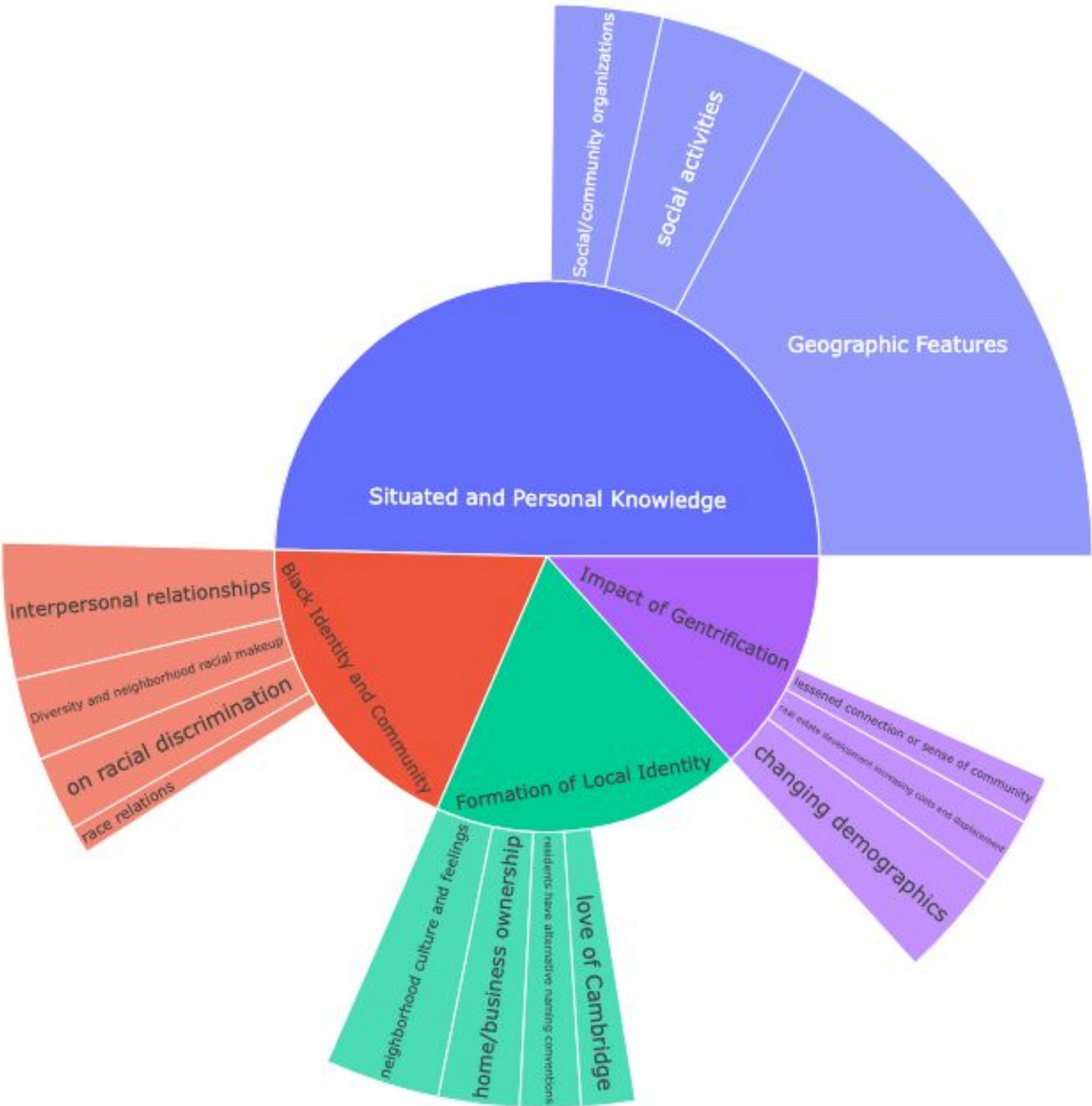
Names of churches, schools, streets, and other activity spaces to serve as anchors around which to orient more situated knowledge

Pride in Community and Homeownership

All women spoke with pride, affection, and nostalgia for their neighborhoods, particularly about homeownership and communal care

Prevalent Negativity about Gentrification

Despite conflicting opinions on race in the 20th century, all women mentioned demographic, social and cultural changes resulting from displacement caused by real estate speculation



"expressive power of geospatial technologies"

"GIS can integrate a wide variety of geographic data and qualitative materials and can provide a multimedia interactive environment for... telling stories about people's lived experiences"

"illustrate how oral histories may be used to construct GIS-based visual narratives"

highlighting marginalized perspectives and forms of knowledge

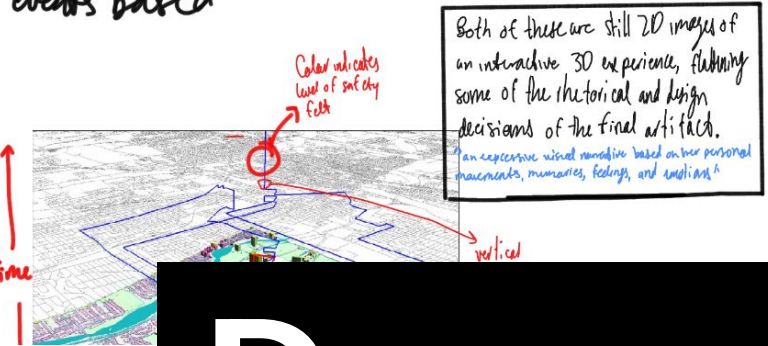
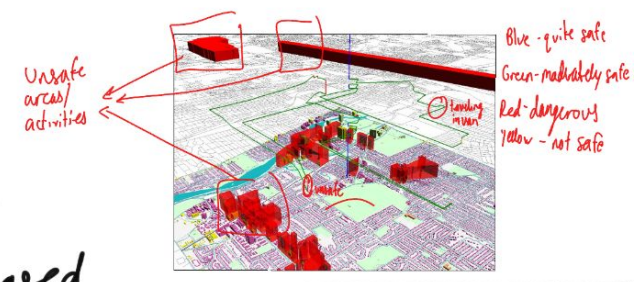
Oral Histories + Diary Data + GIS → visual narrative

"a life path" that traces the temporal sequence of events based on a person's activities and trips

Central narrative device

Background data on buildings, rivers, streets, etc.

arguments in critical/feminist GIS



Documentation

7/23/26 - Keywords

Houghton Elementary School	Born on Pine Street	Peabody Grammar School	Dimmock St, Boston. New England Hospital for Women and Children
The Coast	Lived on Windsor Street	"people owned their own homes"	211 Hovey Avenue, 103 Inman Street, 306 Wenden Street
"the city designates as River's side"	Roberts School	Copley Street	Peabody on Linnaean St
Cambridge Community Center	Margaret Fuller House - "the place to be"	Fayerweather Street	dwindling Black community
Regina Key	Cherry Street Neighborhood House	Saint Bartholomew's Church	Church owned property
216 Green Street	"church was our entertainment"	"I didn't feel anything on Harvard St"	"always welcome"
Pearl and Green St	Barbados	"racially different"	"owned these houses"
"cost of living"	"A lot of people from Barbados in Cambridge"	Filing's Basement	"corner of..."
"the demographics have changed tremendously"	First Helena's Church	"I don't have that connection anymore"	ice skating
Illness resource allocation	Mrs. Nichols	North Cambridge (technically west Cambridge)	Concord Avenue
advocacy service	St. Bartholomew's Church	House parties	Union Baptist Church
"identified... by our politicians"	Concord Ave	Chicken salad sandwiches	Mass Ave Baptist Church on Hampshire Street
Margaret Fuller House "was once very diverse"	Community Barbershop	The Breakfast Club	"knew the neighbors"
The Port/ Area 4 (Federal designation)	Granville Road	Cambridge Women's Community club	"you got to know people"
Inner Belt	Fayerweather Street	"Not directly and not blatant" racism	Peabody Grammar School
	"once you cross West End Lane, nobody was white"	"subtle" "microaggression"	Linnaean Street
	"turned into West Cambridge"	"you found people through whatever church you went to"	walking to/from school
	The Russel School	Service	Porter Square



Corresponding Codes

home/business ownership community orgs

namng conventions

interpersonal relationships

social activities

love of cambridge

social activities

lessened sense of community

changing demographics

real estate costs and displacement

race and race relations

Mapped streets, room spaces (churches, schools, community centers), and leisure spaces on 3 layers.

This is my guidance system, allowing me to see where on the map emotions were felt.

Limited by the GIS online

- animations are time-based, used opacity instead
- learning curve makes this wholly inaccessible

Themes → Layers

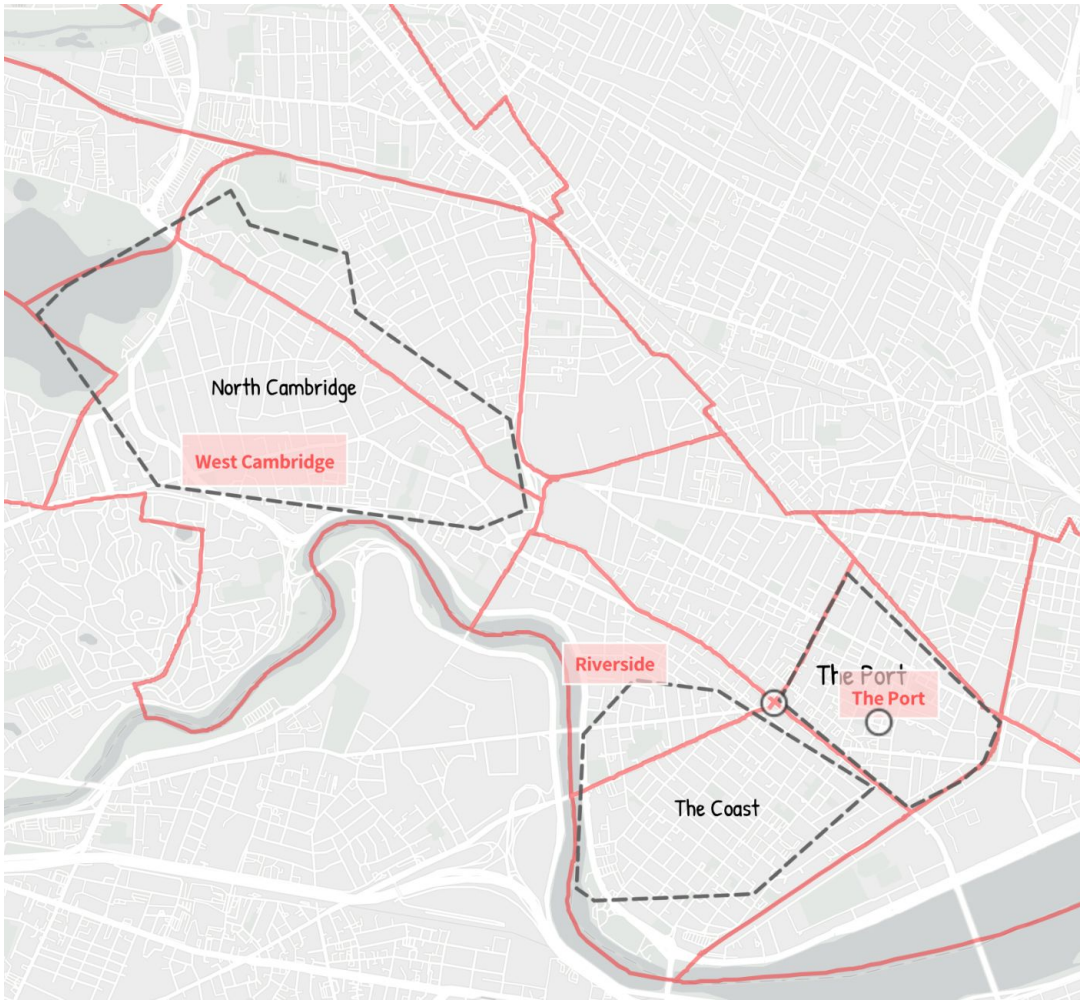
Black Identity + Community → Race and Race Relations

Formation of Local Identity → Local Pride and Identity

Situated And personal knowledge

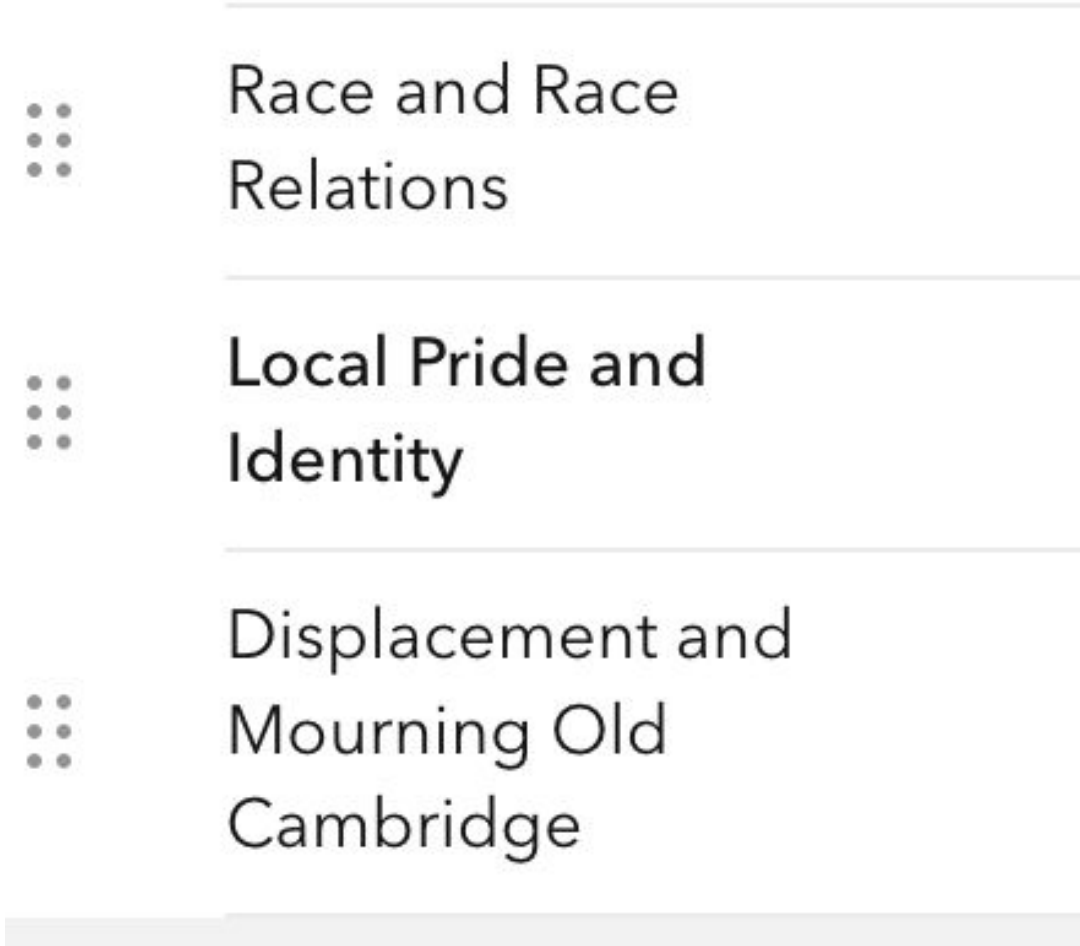
Findings from Final Visualization

Self-Identification vs Imposed Definition



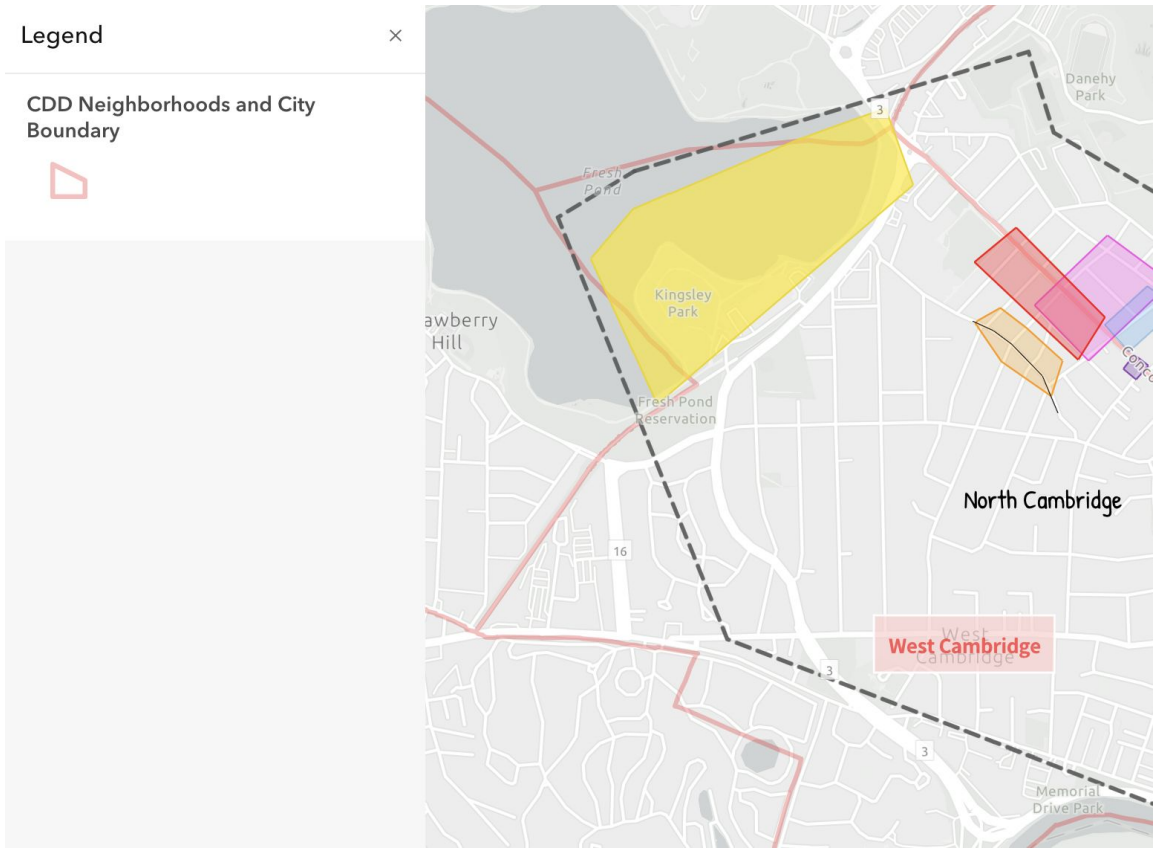
Neighborhoods were named and defined differently by these Black women than by the city. **Centering these borders and endonyms** counters the definitions imposed by city data and reinforced by ArcGIS basemaps

Emotional Complexity through Layers



Despite feelings of disconnection and frustration with gentrification, these women still discussed their homes and neighborhoods with pride and affection. **Map layers can reveal, hide, and overlay these emotions** to convey the complexity of these women’s experiences

Minimal Possible Disruption



Throughout the design process, I ran into limitations with including sketch features in the map legend and stylizing pop-ups that were expected given the attempt to translate qualitative data in a quantitative platform.

Conclusion

Implications

- ***Cultural Memory in Cambridge:*** These mapping the embodied and emotional geographies of Black Cantabrigian women offers unique insights on how the continued development of the city directly impacts.
- ***Critical GIS:*** This work corroborates previous findings in critical GIS that identify limitations of other software in visualizing embodied geographies and specifically explores these limitations as they apply to Black women's geographies.

Future Work

- ***Larger Scale/Extent:*** Examining Black Women's geographies across multiple cities or diasporic perspectives, or with a larger corpus of oral histories offers more variation and depth of knowledge
- ***Participatory Approach:*** Semi-structured interviews specifically about Black women's geographic perspectives and/or community mapping projects provide situated knowledge from primary sources

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